Big Data
(MEI)

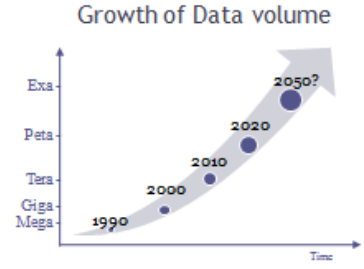
João Pedro Costa
jcosta@isec.pt

Agenda

- O aumento de dados
  - Maior número de ficheiros
  - Ficheiros maiores
- Problemas ..
  - Como e onde guardar?
  - Como processar?
- google research
  - Google File system
  - Paradigma Map-Reduce

Data Volume

- Increasing
- Imagine the volume



How to handle this?

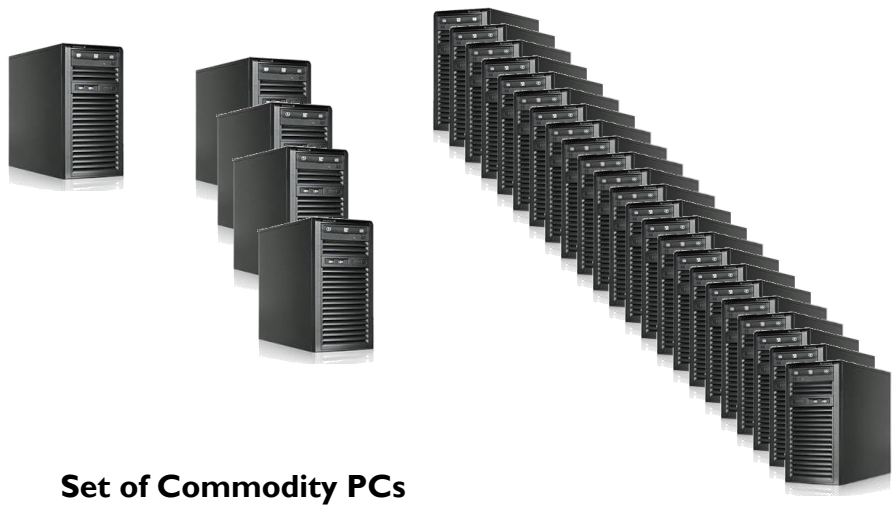
- Infrastructure
  - Scale up
  - Scale out
  - Distributed processing infrastructures
- Storage
  - How and where to store?
  - How processing nodes can access it
- Computation
  - Parallel processing
  - Distributed processing

Infrastructure – Scale up



High-end HW

Infrastructure – Scale out



Set of Commodity PCs

Infrastructure



Storage

- Where data is stored?
- How data is stored?

Where data is stored?



Issues :
- Capacity
- Throughput
- Reliability
- Cost

The World's First Disk Drive: RAMAC 350
It has 50 24" disks spinning at 1,200 RPM holding 5 million characters of information.

Where data is stored?



Seagate BarraCuda 8TB Internal Hard Drive HDD – 3.5 Inch Sata 6 Gb/s 5400 RPM 256MB Cache for Computer Desktop PC – Frustration Free Packaging (ST8000DM004)
Visit the Seagate Store
55,857 ratings
767 answered questions
Price: $139.99 + $15.56 shipping



Seagate 4 TB Terascale HDD SATA 6Gb/s 64MB Cache 3.5-Inch Internal Bare Drive (ST4000NC001) (Renewed)
Visit the Amazon Renewed Store
60 ratings | 4 answered questions
New Price: $63.88
Price: $53.99 + $30.31 Shipping & Import Fees Deposit to Portugal Details
You Save: $9.89 (15%)
Style: Standard Model



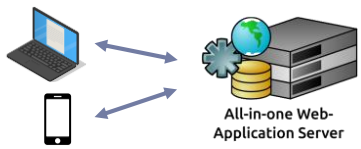
HP Storeeasy 1630 28TB SAS Storage (B7D95A)
Visit the HP Store
Price: $10,795.00
Available at a lower price from other sellers that ma not offer free Prime shipping.
- NAS Server
- 28TB HD capacity
- 7,200 rpm

Used (3) from $7,595.20

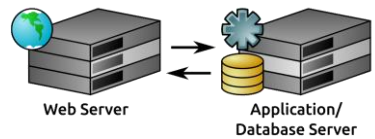
Where data is stored?

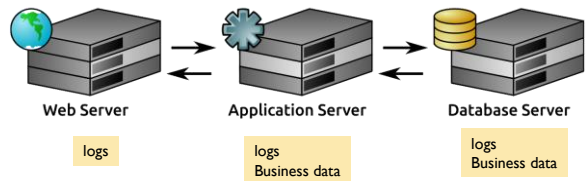


Collecting data …

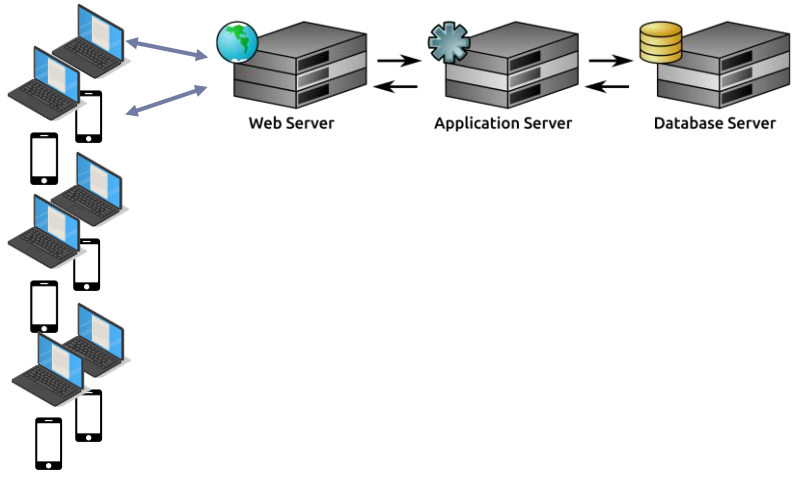


Business data
Log data

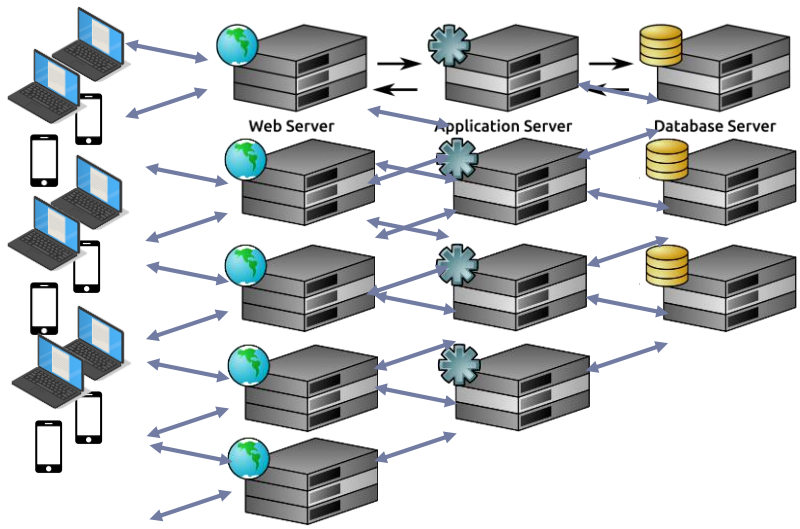




Scalability



Scalability

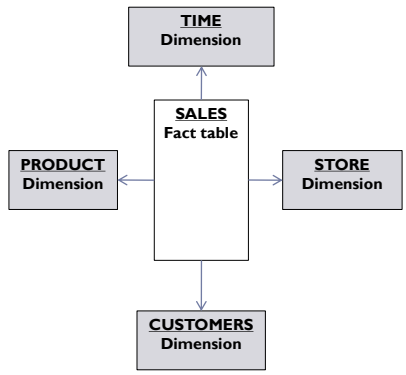


Scalability - Issues

- Partition
  - How ? What ? Where? When ?
- Replication
  - How ? What ? Where? When?
- Consistency:
  - Strong
  - Weak
  - How ?
- Data Access / Pattern
  - Query one
  - Query subset
  - Insert/update

Partition among several nodes.
- What ?
- How ?
- Where?
- When ?

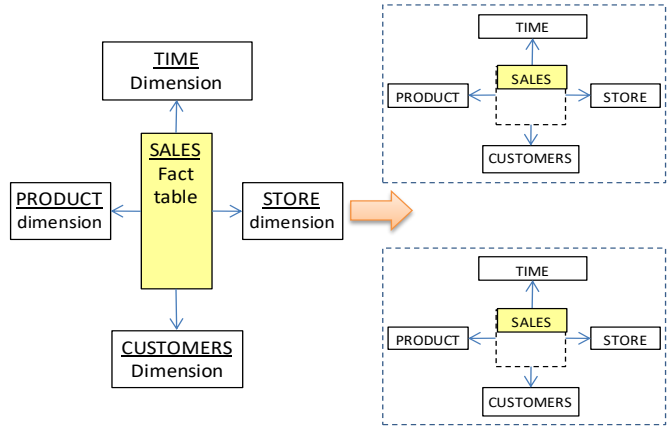
How a RDBMS handle a large DW?



Using parallel RDBMS →
Partition among several nodes.
- What ?
- How ?
- Where?
- When ?

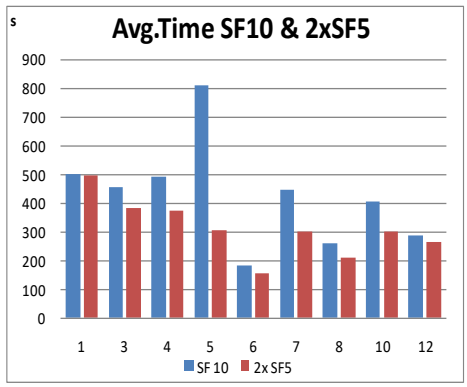
How a RDBMS can handle a large DW?

Fact table partitioned into 2 nodes



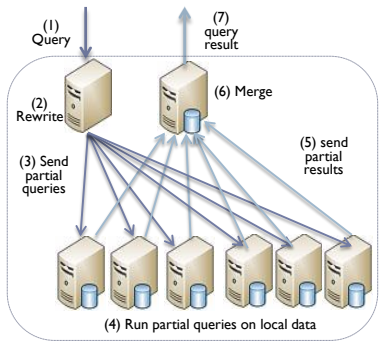
How a RDBMS can handle a large DW?

Fact table partitioned into 2 nodes



10GB vs 2x 5GB.
Doesn't take into account
- Merging overhead
- Data skew
- Data distribution

Parallel Query Processing



Execution time influenced by

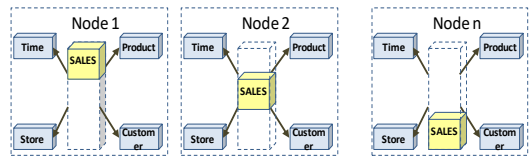
- Query selectivity
- Data volume in each node
- Execution plan
- Join algorithm (local vs in-network)
- Join order
- Nodes' characteristics

Contrained by the slowest node

Execution time

$$t_{execTime} = t_{rewrite}(Q) + t_{send}(Q) + \max\left(t_{Nexec}(Q) + t_{Nsend}(Q_{pR}) + t_{receive}(Q_{pR})\right) + t_{merge}\left(\sum Q_{pR}\right) + t_{send}(Q_{fR})$$

How a RDBMS can handle a large DW?



Issues

- How to distribute data among nodes?
  - Replicate dimensions? But dimensions can be large
  - Equi-partitioning? Which dimensions? May require repartitioning
- How to Handle Nodes Heterogeneity?
- How to balance data with the nodes' performance?
- How to dimension the parallel infrastructure?
- How to handle data heterogeneity?